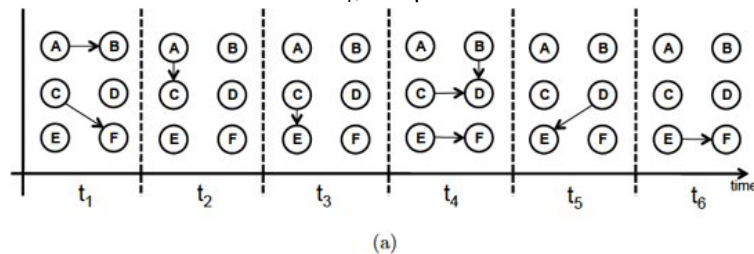


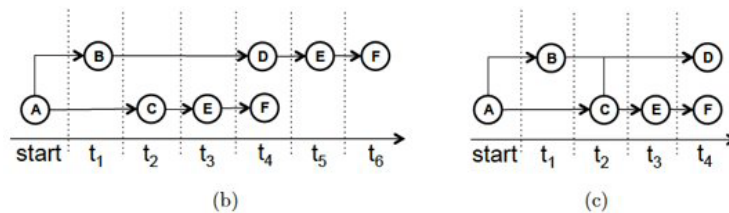
L14: Advanced Topics: Temporal Networks

An active research area in Network Science is the study of *Temporal Networks*. In such networks, nodes and edges may be present only for specific time periods. For instance, a network that shows the phone communications between people may represent each call with an edge that has a specific start and end-time. Note that the transitivity property does not apply in temporal networks, in the sense that if node-A connects to node-B, and node-B connects to node-C, we cannot conclude that node-A connects to node-C.

For instance, Figure-a represents the communications between six people, arranged temporally in six time steps. For instance, during the first time period (t_1) node-A contacted node-B, and node-C contacted node-F. If we ignore the temporal ordering of the links, we can reach completely wrong conclusions about the communication flows that are possible in a given network. For instance, if node-A is the only node with some information at time t_1 , is it possible that this information will ever reach node-F



during the following six time steps?



Source: "[Temporal network metrics and their application to real world networks](https://www.cl.cam.ac.uk/~cm542/phds/johntang.pdf)" (https://www.cl.cam.ac.uk/~cm542/phds/johntang.pdf) (Figure 4.5) by J.K.Tang.

Figure-b shows that there are actually two different "*temporal paths*" from node-A to node-F. In one of them node-F would obtain the information at time t_4 , while in the other at t_6 .

We could also ask: what is the earliest time at which each node in the graph could receive this information? This is shown in Figure-c, referred to as the "*minimum temporal spanning tree*" originating at node-A.

These are only a couple of the many interesting questions we can ask about temporal networks. The temporal structure of edge activations can significantly affect the dynamics of processes that are taking place on a network, such as epidemics, information diffusion, or synchronization. Additionally, the metrics we have defined earlier to quantify the centrality of nodes (or groups of nodes) have to be modified in temporal networks so that they only consider temporally-valid paths.

If you are interested to learn more about Temporal Networks, we recommend the recent book "[A Guide to Temporal Networks](https://www.worldscientific.com/worldscibooks/10.1142/q0033)" (https://www.worldscientific.com/worldscibooks/10.1142/q0033) by R.Lambiotte and N.Masuda.